Eduardo Gaitán Escalante

Data Scientist & AI Engineer · Economist · Freelance / project work, 10-20 h/week (UTC-6, Mexico)

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SUMMARY

Economist (UNAM, honors) working as a data scientist and AI engineer. I build forecasting, econometric and machine-learning models that answer concrete business questions, then ship them as working systems: data pipelines, LLM-powered automation, dashboards and containerized services on cloud or VPS. Freelance and contract work for small and mid-size companies in Mexico and for remote clients; deliverables are reproducible code, a written report and the system running in production.

TECHNICAL SKILLS

Machine learning & statistics: Python and R (advanced); scikit-learn, XGBoost, LightGBM, H2O, statsmodels, Isolation Forest, K-Means/DBSCAN; forecasting (Prophet, SARIMA, ETS, Croston), spatial econometrics, held-out and spatial-block validation, Monte Carlo/Sobol. PyTorch/TensorFlow for satellite imagery.

AI engineering: Anthropic Claude API with structured outputs, local models with Ollama (Qwen3-VL), Gemini, n8n, REST APIs, webhooks, OAuth2, schema validation with Pydantic/Zod, per-call cost control and human-approval gates.

Data engineering & cloud: SQL, PostgreSQL, SQLite, BigQuery, Polars, pandas, Dask, Parquet, Kedro, MLflow, ETL; Google Cloud (Cloud Run, Cloud Build, Cloud Storage), Spark/PySpark and Databricks (certifications and one experiment), web scraping with Playwright.

Analytics & reporting: Power BI, Streamlit, Plotly, MapLibre/PMTiles, GeoPandas; R tidyverse, ggplot2, sf and R Markdown; Excel/openpyxl and python-pptx financial reporting; INEGI/CONEVAL/NASA data.

Applications & delivery: TypeScript/JavaScript, React, Next.js, Astro, Tailwind CSS, Vite, FastAPI/Flask, Node.js/Fastify/Express, Stripe/Openpay, Telegram/WhatsApp; Docker Compose, Linux, Traefik/nginx, GitHub Actions, pytest, Vitest, Playwright, Trivy and CrowdSec. Project-specific: C#/.NET 8 Windows print agent with xUnit tests.

EXPERIENCE

Independent Data Scientist & AI Engineer (freelance / contract) — Jan 2025 - Present

- Market, demand and pricing studies for small businesses: Huff gravity demand model validated out of sample, spatial econometric models with spatial-block cross-validation, competition clustering, ETS/SARIMA forecasts, price-elasticity and seasonality analysis, and Monte Carlo capital planning with Sobol sensitivity; delivered as reproducible Python reports and investor decks.
- With SmartCFO AI (data and finance consultancy): per-SKU, per-branch demand-forecasting pipelines (Kedro, Prophet, SARIMA, XGBoost, Croston, Mexican-calendar features) tracked in MLflow on BigQuery and Cloud Run; trip-level cost and revenue analytics; revenue-opportunity dashboards. Contributed to Smart CFO's SaaS MVP for business valuation and financial projections.
- LLM automation in production: a receipt-reading accounting bot on the Claude API with structured outputs (about US$0.02 per receipt) and a local Qwen3-VL fallback; a Claude-based incident analyst for server monitoring whose actions require human approval; an agent control plane that runs coding agents, their tests and opens draft pull requests.
- Full systems for restaurants and online stores: point-of-sale and kitchen screens by station, order intake from WhatsApp, web and delivery platforms, Stripe/Openpay payments, hex-grid sales coverage maps and automatic delivery assignment; e-commerce for four brands with live Stripe checkout.
- Operations: Docker Compose and Traefik on a hardened VPS, scripted deploys with SHA-256 verification and rollback, Telegram-based tooling, 1,300+ automated tests on the largest codebase.

Naturgy | Data Science Specialist — Jan 2025 - Jul 2025

- Led a cross-functional team in designing and deploying an **Isolation Forest** model for anomaly detection, including ETL pipelines, automation workflows, orchestration, hyperparameter tuning, and performance analysis using H2O and R.
- Modernized a legacy **K-Means** customer-segmentation model, achieving a 15% improvement in accuracy and a 10% reduction in training time.
- Partnered with Economics and Finance to automate 7 manual processes, reducing processing time from 5 hours per task to under 1 minute and saving hundreds of hours annually.
- Proposed and developed a hybrid boosted forecasting model (XGBoost + seasonality) for annual natural-gas consumption.

National Council of Evaluation of the Social Development Policy | Head of the Small Area Estimation Methodologies Department — Jan 2024 - Jan 2025

- Led a data-pipeline project using Docker, Kedro, PostgreSQL, pgAdmin4, and Git, improving storage and analysis workflows.
- Fitted supervised and unsupervised models including Random Forest, XG/Gradient Boosting, Regression/Logistic, SVM, neural networks, and hybrid boosted time-series models.
- Used SQL to manage queries and manipulate datasets efficiently in pgAdmin4; contributed to poverty-analysis publications within the technical estimation team.

General Coordination for Planning and Investment - JGCDMX | Environment, Tourism, and Economic Development Coordinator Oct - Nov 2023

- Automated ETL processes with Python scripts, reducing project-creation times by 20% and making information available to land teams sooner.
- Developed interactive geospatial dashboards using GeoPandas, Leaflet, and Folium for decision-making and route planning.
- Deployed two geospatial models (K-means and Spatial Lag Model) for heatmaps, analytical reporting, and pattern distribution.

Python & R Developer | Program for Development Studies - UNAM Jan 2020 - Sep 2023

- Led the creation of a geospatial night-time-light data pipeline, optimizing analysis, storage, and reporting with Parquet.
- Designed Docker containers for developer environments integrating PostgreSQL databases, Python, and R tools.
- Conducted advanced geospatial estimations for socioeconomic research purposes.

SELECTED PROJECTS (2026)

Demand, competition and capital study for a dark kitchen (client confidential) Sep - Oct 2026

- Huff gravity model calibrated with leave-one-municipality-out validation (median absolute error 23% vs 74% for a naive baseline); OLS/SAR/SEM models with spatial-block CV; DBSCAN/KDE competition clustering on INEGI DENUE; damped ETS vs SARIMA forecasts; 20,000-run Monte Carlo with Sobol sensitivity for the capital plan. Python, pandas, LightGBM, ENIGH/Census data, Sentinel-2 CNN features.

Point-of-sale and kitchen-display platform for a dark kitchen | bearandchimps.com Jul - Oct 2026

- Fastify + SQLite + React/TypeScript system with kitchen, packing and delivery station screens, web and delivery-platform order intake via signed webhooks, card payments, Telegram operations bots and an LLM receipt reader (Claude API with a local Qwen3-VL fallback); 1,300+ automated tests, scripted deploys with rollback.

Housing price model and data pitch for developers, Monterrey Aug 2026

- Playwright scrape of public listings (9,653 published, 936 clean houses across 12 municipalities); hedonic regression (n=791, R^2 0.85, 19% median out-of-sample error, 5-fold CV) with interpretable coefficients per bathroom, land, parking and bedroom; 17-slide deck and PDF.

Demand-forecasting pipeline for a distributor | with SmartCFO AI Sep 2025 - May 2026

- Kedro pipeline (Prophet, SARIMA, XGBoost, Croston/SBC) over ~4,940 branch × manufacturer series, 52-week horizon with 4-week validation, MLflow tracking, BigQuery and Cloud Build; 93× speed-up from vectorization and a documented fix for a flat-forecast regression.

Yoga Verde: online store and market study | yogaverdemx.com Aug - Oct 2026

- Astro storefront with a private Stripe Checkout backend and SQLite inventory; reproducible study of unit economics by channel, price elasticity, seasonality and INEGI income geography, published as an interactive report.

Tide: open-source, offline-first cycle tracker | github.com/eduardoge13/tide-cycle-tracker Oct 2026

- React/TypeScript PWA with no backend and a strict CSP; statistical prediction engine (median for short histories, recency-weighted mean for 6+ cycles, confidence rules), ovulation estimate learned from LH tests, Vitest suite, CI and GitHub Pages deploy.

ADDITIONAL PROJECTS & PUBLIC CODE

Night-time lights and poverty atlas | github.com/eduardoge13/luces-noche-pobreza 2023 - 2026

- NASA VIIRS Black Marble vs CONEVAL poverty for 2,469 municipalities (2019-2024) for an academic book chapter; SLX, spatial-lag and spatial-error models; Next.js + MapLibre + PMTiles atlas whose audit documented three errors in the original analysis.

Punto Clave MX e-commerce | puntoclaveshop.com.mx 2026

- Next.js 16 storefront with 3D showroom, live Stripe Checkout with 3-D Secure, SPEI, chargeback hardening and GitHub Actions deploys.

Payment reminder system | github.com/eduardoge13/payments-reminder 2025

- RFM segmentation, ML-based channel selection, WhatsApp/Twilio messaging, Notion and scheduled GitHub Actions.

Databricks experiment | MLflow Dec 2024

- Gradient Boosting classifier for air-quality labels with MLflow experiment tracking.

EDUCATION & CERTIFICATIONS

Universidad Nacional Autónoma de México | B.A. in Economics, graduated with honors (GPA 9.4/10) Oct 2024

- Coursework: Econometrics, Statistics, Time Series, Mathematics, Linear Algebra, Game Theory, Economic Development.
- Certifications: Machine Learning with Apache Spark & PySpark; Databricks Fundamentals; Geographic Data Science with Python (Center for Spatial Data Science, University of Chicago); Intro to Deep Learning & Time Series (Kaggle); Learning Amazon SageMaker (LinkedIn Learning); Unlock Your Data with Data Cloud (Salesforce).